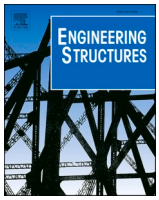
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Metaheuristic-optimized vision–language framework for quantitative assessment of concrete bridge deterioration

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ABSTRACT

Timely and reliable assessment of concrete deterioration is critical for ensuring the safety and serviceability of bridge structures, yet conventional visual inspections remain labor-intensive, subjective, and difficult to standardize. Current artificial intelligence (AI)-based methods often lack the fine-grained quantification and domain-specific reporting required for reliable cost estimation. This study proposes a non-destructive evaluation framework that integrates computer vision with multimodal large language models (MLLMs) to automate the detection, quantification, and documentation of concrete bridge defects. A You Only Look Once (YOLO)11-seg instance segmentation model, fine-tuned using the Pilgrimage Walk Optimization (PWO) algorithm, identifies and classifies five concrete deterioration types. Pixel-level masks are further used to quantify defect areas for condition assessment and repair cost estimation. In parallel, a Retrieval-Augmented Generation (RAG)–enhanced language model, fine-tuned with Low-Rank Adaptation (LoRA) for parameter-efficient training, generates component-specific inspection descriptions and standardized repair recommendations in accordance with engineering guidelines. Experimental results demonstrate that PWO enhances segmentation performance compared with baseline training, and Bidirectional Encoder Representations from Transformers Score (BERTScore) analysis confirms strong semantic alignment between generated and reference reports. By coupling metaheuristic optimization with retrieval-grounded language generation, the framework enhances the precision, consistency, and reproducibility of concrete deterioration evaluation. The proposed framework provides a practical tool for automated condition evaluation, engineering documentation, and maintenance decision-making in bridge inspection workflows.

1. Introduction

Labor shortages have worsened across labor-intensive sectors, especially construction. According to the Ministry of Labor's 2024 first-quarter manpower survey, demand for construction engineers grew by 3.69% year-over-year to 4276 roles—the highest number recorded [1]. Compared with 2015, this represents a 552.1% increase, highlighting ongoing workforce gaps. Analyses by the McKinsey Global Institute [2] and the Taiwan Construction Research Institute also show that productivity in the construction industry has stayed near 1% annually, significantly below global economic and manufacturing standards. Although construction generates about USD 10 trillion annually, stagnant productivity emphasizes the need for digital solutions to enhance efficiency and reliability in inspections and maintenance, where manual work remains prevalent.

Bridge structures form a critical part of Taiwan's transportation network. Among the approximately 26,000 bridges nationwide [3], more than one-third have exceeded 30 years of service, raising concern about age-related deterioration of concrete components. Current inspection practice relies heavily on manual visual surveys, followed by written documentation [4,5]. These workflows are time-consuming and labor-intensive, and often constrained by access conditions, leading to variability in defect identification and delays in maintenance planning. Reports in 2022 noted ongoing shortages of inspection personnel and funding, further affecting the frequency and consistency of routine assessments. In this context, efficient, accurate, and scalable non-destructive evaluation (NDE) technologies are essential for ensuring structural safety and supporting timely repair decisions [6–8].

When inspections are delayed or findings are inconsistent, maintenance prioritization becomes less reliable. Studies have shown that

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0141-0296/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

condition ratings can vary by up to 5 levels among inspectors, with only 68% falling within ±1 point of the mean [9]. As a result, interventions are often postponed until defects propagate and require more extensive repairs. Research indicates that each dollar of deferred maintenance may result in four dollars or more in future capital renewal costs. Bridge-level analyses further suggest that 10-year maintenance delays can increase total discounted costs by 1.5–2.5 times relative to optimal policies [10]. Such delays extend maintenance lead times and increase structural risk exposure, particularly for aging bridges with progressive deterioration. By providing repeatable defect quantification and standardized documentation, automated systems can facilitate earlier intervention, improve condition-based prioritization, and enhance the consistency of safety-related decision-making.

Recent advances in artificial intelligence, particularly Generative AI and multimodal large language models (MLLMs), have expanded opportunities for automating concrete condition assessment [11,12]. Although computer vision techniques have been used to detect visible deterioration in bridge imagery, many existing approaches focus on image-level classification and seldom integrate language models capable of producing standardized engineering documentation. As MLLMs evolve, joint modeling of visual and textual information enables the automated generation of inspection narratives aligned with domain-specific terminology, thereby promoting greater consistency in defect descriptions and repair recommendations. These capabilities reduce reliance on manual report writing and enhance reproducibility, throughput, and overall documentation quality.

To address these limitations, this research introduces a multimodal framework that transcends conventional automated inspection. Unlike prior studies that treat image classification and report writing as decoupled tasks, our approach integrates visual semantics and engineering logic. By fine-tuning MLLMs with domain-specific knowledge, the proposed method not only detects deterioration but also interprets its significance within specific structural components. For instance, it can distinguish the structural implications of a crack in a pier from those in a deck slab. Furthermore, the framework bridges the gap between technical assessment and asset management by linking detected defect severities directly to quantitative unit repair costs. This synthesis of vision-based detection, standardized narrative generation, and economic quantification establishes a cohesive, actionable workflow that mitigates the subjectivity of manual surveys and the information silos of traditional AI tools.

This study focuses on concrete bridges, which constitute the majority of the regional inventory [13]. The component taxonomy is derived from municipal safety inspection guidelines [14] and targets 10 representative structural elements (e.g., piers, girders, and deck slabs). The assessment framework covers five prevalent deterioration categories identified in the literature [15,16]: cracking, spalling, honeycombing, exposed reinforcement, and vegetation. Data sources include municipal inspection records, field surveys, and the public CODEBRIM dataset [17]. Fig. 1 illustrates the overall architecture of the proposed metaheuristic-optimized vision–language framework. The system adopts a dual-stream structure comprising a vision module for defect localization and quantitative measurement and a language module for structured narrative generation. Both modules are optimized through a PWO-based hyperparameter tuning strategy. The outputs from the two streams are integrated via a decision-level fusion mechanism, linking pixel-level defect masks to area-based cost estimation and standardized inspection report generation.

2. Literature review

This study develops an instance-segmentation model combined with an MLLM to identify concrete bridge deterioration and generate automated visual inspection reports. The model's accuracy and reliability are validated through quantitative evaluations and expert surveys, thereby reducing time and labor while providing government agencies and inspection organizations with valuable insights into visual inspection practices and repair cost estimates. This section (i) reviews the evolution of bridge inspection and the use of Artificial Intelligence (AI) in engineering inspections, (ii) explains the technical foundation of MLLMs and instance segmentation for future system development, and (iii) examines optimization algorithms in deep learning for construction, highlighting the practical relevance of the chosen approach.

2.1. Evolution of bridge inspection and AI in engineering practice

Bridge inspection is fundamental to ensuring structural safety. Traditional methods include visual inspection, destructive testing, and non-destructive testing. This study emphasizes visual inspection, which still relies heavily on on-site evaluations and image collection by inspectors, followed by report compilation. This process is time-consuming, labor-intensive, and difficult to scale. In recent years, AI has been increasingly applied to bridge inspection and related engineering tasks, including deterioration assessment and cost estimation.

Kunlamai et al. [18] were among the first to integrate bridge inspection with Visual Question Answering (VQA). They pre-trained a VQA model on image–text pairs from inspection reports, enabling it to identify and briefly describe defects such as cracks, corrosion, deformation, and exposed rebar. Building on this direction, Chun et al. [19] developed a deep learning image captioning approach using

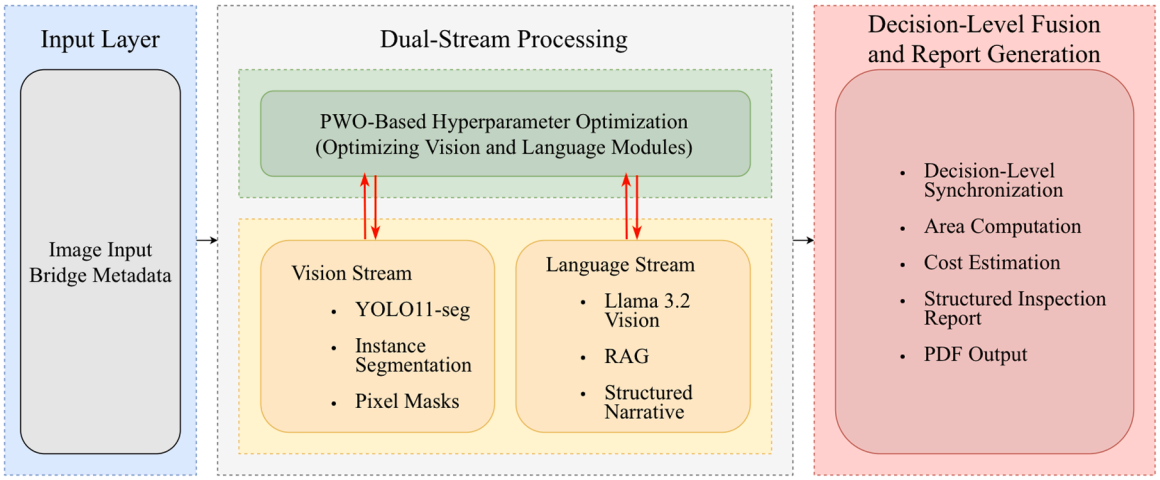


Fig. 1. Architecture of the proposed metaheuristic-optimized vision–language framework.

convolutional neural networks (CNNs) for feature extraction and long short-term memory (LSTM) networks for sequence generation. Their system automatically generated descriptive text for uploaded bridge images, which could be refined via a feedback mechanism to improve model performance.

Chou et al. [16] proposed a deterioration recognition framework for bridge backwalls and substructures by combining a Vision Transformer (ViT) classifier with the YOLOv8 instance segmentation model. This system classified major components, detected detailed deterioration patterns, and calculated deterioration areas through polygon-based segmentation. The quantified damage was then combined with unit maintenance costs to estimate repair expenses.

In parallel, Pu et al. [20] introduced an automated construction-reporting framework that leverages a multimodal large language model. Scene data collected via autonomous vehicles were reconstructed into 3D models and paired with manual reports to fine-tune the MiniGPT4–7B multimodal model. The resulting system automatically generated draft construction reports.

Collectively, prior studies demonstrate significant progress in applying computer vision and multimodal learning to bridge inspection. However, existing research tends to focus on individual components of the inspection workflow rather than their systematic integration. Many studies focus on defect detection or classification using convolutional- or transformer-based vision models, while others explore automated text generation using vision–language architectures. Although some frameworks incorporate quantitative estimation or cost calculation, these components are rarely combined with pixel-level instance segmentation and domain-specific standardized reporting within a unified pipeline. As a result, the entire technical process, which includes accurately locating defects, quantifying measurable deterioration, and producing structured inspection documentation, has not been fully addressed in the existing literature.

2.2. Multimodal language models and instance segmentation for bridge assessment

Recent advances in MLLMs and instance segmentation networks have opened new opportunities to automate the assessment of bridge deterioration. In safety-critical civil infrastructure, two requirements are paramount: (i) fine-grained localization of minor, low-contrast defects and (ii) generation of standardized, engineer-readable text aligned with inspection taxonomies and maintenance practices. Research streams in lightweight vision–language systems, mid-scale unified models, pixel-level multimodal architectures, and high-throughput segmentation networks provide partial support for these needs but remain incomplete.

Alternative models such as Florence-2 [21], PaliGemma [22], and Vitron [23] offer functional multimodal capabilities but lack fine-grained defect mapping and standardized reporting. In contrast, this study employs Llama 3.2 Vision for domain-adapted reporting and YOLO11-seg for instance segmentation, both selected for their openness, fine-tuning flexibility, reliable detection of subtle defects, and reproducibility in deployment. The modular pairing enables independent metaheuristic optimization of the vision and language components, with optional Retrieval-Augmented Generation (RAG) to enrich repair guidance. Together, these choices support the dual objectives of this work: generating precise pixel-level masks for quantity takeoffs and cost estimation, and producing standardized narratives consistent with inspection taxonomies and maintenance standards.

Launched in September 2024, Meta's Llama 3.2 series includes four model sizes: 1B and 3B (text-only) and 11B and 90B (Llama 3.2 Vision), which jointly process images and text. These Vision models extend the Llama architecture by adopting a decoder-only architecture optimized for language generation. At the same time, the vision encoder converts images into token sequences, which are then processed in a unified multimodal space. Unlike earlier concatenative schemes, Llama 3.2 Vision integrates visual and textual representations directly, enabling consistent semantics and coherent reasoning, making it a practical option for vision–language integration in deployment-oriented engineering workflows.

Meanwhile, Ultralytics introduced YOLO11 in October 2024 as a real-time detector with enhanced instance segmentation capabilities. Architectural updates—including replacing C2f blocks with C3K2 blocks and an anchor-free prediction head—improve multi-scale feature representation while maintaining low latency and reduced parameter count. Prior studies have demonstrated the strong performance of YOLO variants in bridge inspection, ranging from crack detection to general deterioration analysis [24–27]. Building on these findings, YOLO11-seg offers a practical, efficient approach to generating pixel-level defect masks in real-world conditions.

Against this background, the present study advances the research landscape by explicitly integrating instance segmentation, quantitative deterioration measurement, and domain-adapted multimodal report generation into a single inspection framework. Rather than treating detection, measurement, and reporting as separate tasks, this work establishes a coherent end-to-end pipeline that links pixel-level masks to numerical cost estimation and standardized professional narratives. This integration highlights the innovative positioning of the proposed system within the broader context of AI-assisted bridge inspection.

2.3. Optimization algorithms for AI and deep learning in construction

Optimization algorithms play a crucial role in deep learning and AI, significantly enhancing model performance and efficiency [28]. As artificial intelligence and large language models (LLMs) continue to advance, they are reshaping industries across the board. Nevertheless, the complex architectures and vast hyperparameter spaces of LLMs and high-precision computer vision models (such as instance segmentation networks) make training and fine-tuning computationally demanding. Efficient and stable optimization methods are therefore crucial for accelerating AI development. Traditionally, hyperparameter tuning relied on expert knowledge, a time-consuming process limited by human experience. Recent research has instead emphasized algorithmic optimization techniques, highlighting their growing role in modern AI workflows.

The Pilgrimage Walk Optimization (PWO) algorithm is a recent metaheuristic that addresses these challenges [29]. As a gradient-free approach, PWO is well-suited to large-scale, complex optimization problems in which conventional methods often fail. Benchmark tests on standard functions indicate that PWO is a competitive gradient-free optimizer, suggesting its suitability for high-dimensional black-box hyperparameter tuning.

In computer vision, PWO employs a centroid-guided search strategy to optimize model hyperparameters and augmentation settings, thus improving segmentation robustness in complex image backgrounds. This is particularly valuable for detecting bridge deterioration. Prior studies have optimized YOLOv7 and YOLOv8 with PWO [30], achieving notable gains, including a 4.5% improvement in YOLOv8's test-set prediction performance.

PWO has also been extended to hyperparameter tuning for LLMs [31], particularly in domain-specific contexts. For example, in construction procurement, PWO-optimized Taiwan-LLM and Llama3-TAIDE outperformed both non-optimized baselines and general-purpose models (including ChatGPT-4o) in true/false and open-ended consulting tasks. These results demonstrate the practical benefits of metaheuristic optimization for specialized applications. When combined with Low-Rank Adaptation (LoRA), PWO further reduces GPU memory demands during fine-tuning, lowering computational costs and technical barriers.

Collectively, prior studies indicate that PWO is a competitive gradient-free optimizer for hyperparameter tuning in both vision and language models. However, most existing research optimizes these components independently and does not embed them within a unified

inspection framework tailored for bridge deterioration assessment. In contrast, the present study establishes an end-to-end multimodal pipeline that explicitly links pixel-level instance segmentation, quantitative deterioration measurement, and cost estimation, and domain-adapted standardized reporting. By applying a consistent metaheuristic optimization strategy to both the vision and language components, the proposed framework achieves performance gains beyond isolated improvements. It delivers an integrated, practical, deployable solution for AI-assisted bridge inspection. This systematic integration defines the distinctive contribution of this work within the broader research landscape.

3. Methodology

This study proposes a comprehensive dual-stream framework that autonomously transforms visual data into actionable engineering reports, with a focus on automated defect assessment and reporting in construction contexts. The proposed pipeline integrates quantitative defect analysis with qualitative narrative generation through two parallel yet functionally distinct modules: an instance segmentation network (YOLO11-seg) and a multimodal large language model (Llama 3.2 Vision). Input imagery undergoes bifurcated processing in which the vision stream performs defect localization and pixel-level mask generation to enable precise quantification of deterioration areas and estimation of corresponding repair costs. In contrast, the language stream, augmented by Retrieval-Augmented Generation (RAG), interprets visual context to synthesize standardized descriptive narratives and maintenance recommendations. To enable consistent tuning across heterogeneous models under a black-box setting, PWO is adopted as a unified metaheuristic strategy for both the vision and language components. The overall framework thus establishes a coherent integration of visual perception, semantic reasoning, and optimization-driven learning, culminating in an automated pipeline that generates reliable, structured, and decision-oriented engineering reports.

A balance between computational efficiency, data privacy, and engineering precision informs the selection of these specific methodologies. Compared to end-to-end multimodal models like Florence-2 or PaliGemma, the modular coupling of YOLO11-seg and Llama 3.2 Vision offers superior flexibility for domain-specific fine-tuning and local deployment—a critical factor for infrastructure data security. While YOLO11-seg is chosen for its superior trade-off between real-time inference and pixel-level quantification accuracy, the inclusion of RAG addresses the inherent risk of "hallucinations" in large language models by grounding narratives in standardized bridge inspection manuals. Furthermore, PWO is adopted as a unified optimization strategy because it effectively synchronizes the learning process across these heterogeneous vision and language architectures in a black-box setting, offering higher robustness than traditional gradient-based methods for this complex integration. The following sections detail the dataset preparation procedures, the mathematical formulation of the PWO algorithm, the architectural configuration of the dual-stream models, and the final fusion mechanism for report synthesis.

3.1. Multimodal large language model

Llama 3.2 Vision employs an advanced decoder-only Transformer architecture for its reasoning and instruction-following capabilities. The model encodes images as visual tokens and integrates them with textual context via periodic cross-attention. At the same time, gated attention adjusts the weighting of visual evidence to preserve spatial consistency (see Figs. 2 and 3). Crucially, cross-layer feature fusion emphasizes multi-level semantics (e.g., texture and boundaries) without altering the decoder flow. Relative to concatenative multimodal fusion, token-level integration supports semantic alignment and long-range dependency modeling, both of which are needed for component-aware inspection narratives.

3.2. Instance segmentation network

YOLO11-seg is selected as the instance segmentation backbone to

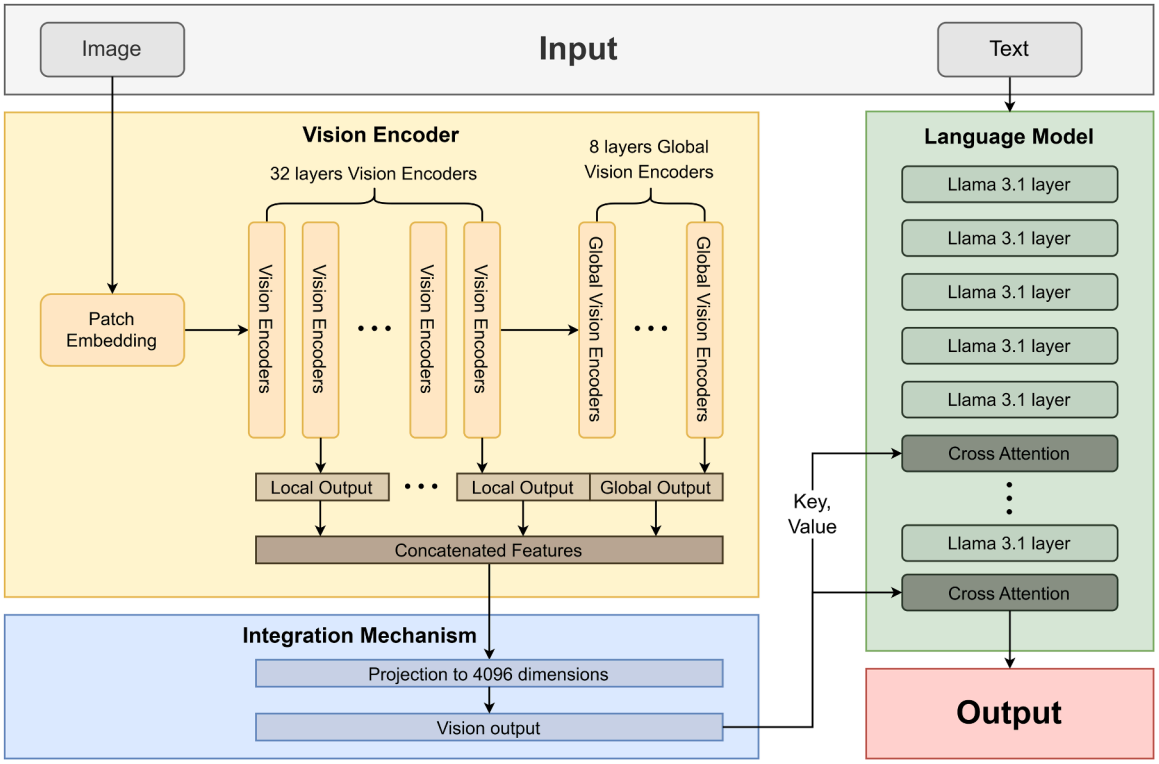


Fig. 2. Architecture of the Llama 3.2 Vision model.

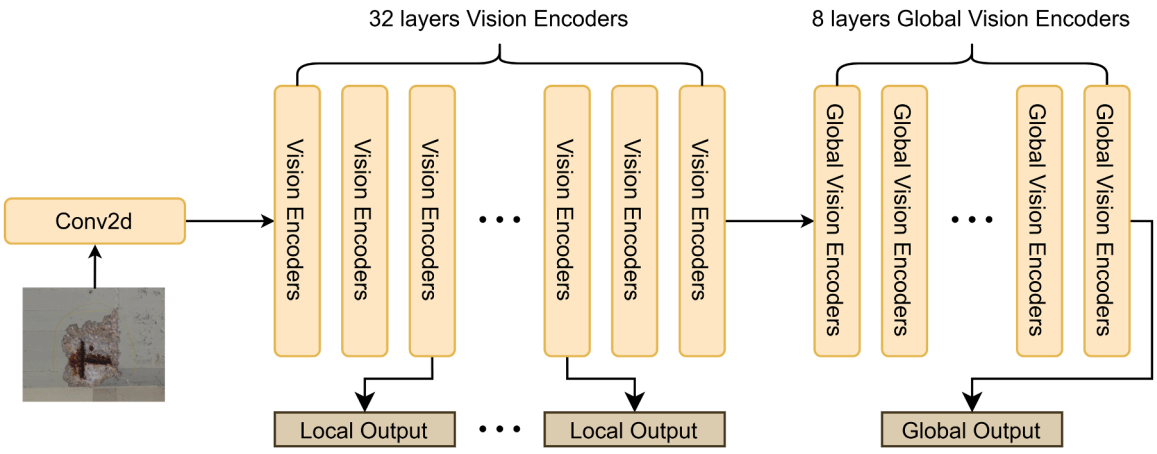


Fig. 3. Illustration of the two-stage feature extraction process.

balance real-time throughput with high accuracy. The architecture introduces key upgrades, including the C3K2 block for efficient feature extraction, Spatial Pyramid Pooling–Fast (SPPF) for multi-scale context, and Cross Stage Partial with Spatial Attention (C2PSA) for saliency enhancement. These features significantly improve cross-scale feature mixing and spatial selectivity, enabling precise detection of subtle, low-contrast deterioration on heterogeneous concrete surfaces (see Fig. 4). The model provides high-quality masks suitable for area-based quantity take-offs while remaining lightweight enough for efficient training and deployment.

3.3. Pilgrimage walk optimization

Pilgrimage Walk Optimization (PWO) is a metaheuristic algorithm introduced by Chou and Liu [29]. Its design is inspired by the cultural significance and collective behavior observed during the traditional Taiwanese religious event, the Mazu Pilgrimage. As illustrated in Fig. 5, the algorithm simulates the congregation and movement patterns of followers accompanying the palanquin along the pilgrimage route. The optimization process is mathematically formalized around these ritualistic stages: (1) divination-block casting, (2) pilgrimage, and (3) return ceremony, complemented by intermediate phases—(A) leisure ceremony, (B) crawling beneath the palanquin, and (C) palanquin robbing.

In the PWO algorithm, each ritual stage corresponds to a specific search operator that regulates population diversity, regional aggregation, intensified local refinement, and termination. This structured mapping establishes a clear connection between the cultural analogy and the underlying exploration–exploitation mechanisms of the optimization process. Together, these stages form a hybrid search strategy that balances global exploration with local exploitation.

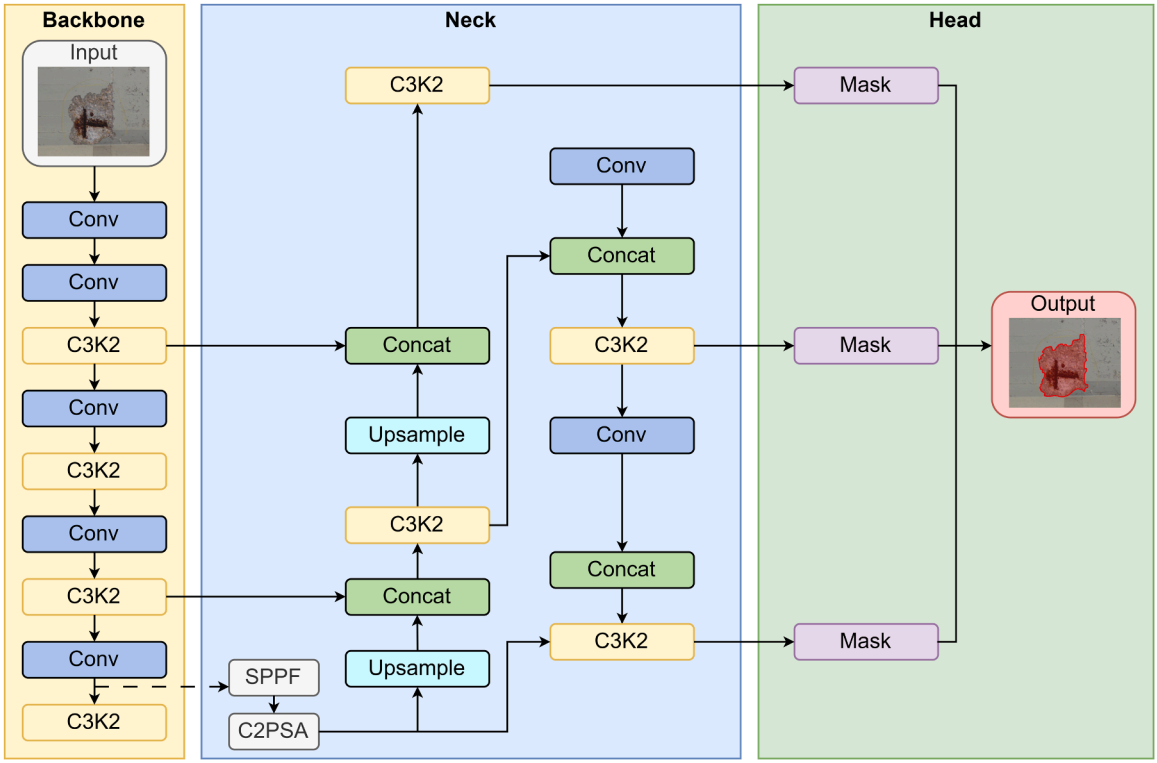


Fig. 4. Architecture of the YOLO11-seg model.



Fig. 5. Conceptual illustration of the Pilgrimage Walk Optimization (PWO) algorithm.

(1) **Divination-block casting.** This stage initializes the particle swarm across the entire search space. This study employs PWO-U, a streamlined variant of the original PWO in which the logistic chaotic map for population initialization is replaced by uniform random sampling. This modification simplifies integration with deep-learning hyperparameter optimization without compromising convergence performance in the hyperparameter spaces considered. The formulation is given in Eq. (1):

$$\mathbf{X}_{\text{inew}} = \mathbf{Lb} + \boldsymbol{r}\cdot(\mathbf{Ub} - \mathbf{Lb}),\quad \boldsymbol{r} \sim \boldsymbol{U}(0,1) \tag{1}$$

where **Lb** and **Ub** represent the lower and upper bounds of the decision variable, and $\boldsymbol{r}$ is a vector of independent random numbers uniformly distributed on the interval [0, 1].

(2) **Pilgrimage.** During the pilgrimage phase, particle aggregation is modeled using K-means clustering to simulate devotees gathering around temples. The population is partitioned into $k = 3$ clusters based on particle positions, after which the clusters are ranked into high-, medium-, and low-fitness groups according to their centroid objective function values. Among the centroids $\{C_1|C_2|C_3\}$, the centroid with the lowest objective function value is denoted as $\mathbf{X}^{\text{best}}_{\text{centroid}}$ and represents the most promising regional search direction at the current iteration.

When $rand \geq 0.5$, particles update their positions toward both the global best $\mathbf{X}^{\text{best}}$ and the best-performing centroid $\mathbf{X}^{\text{best}}_{\text{centroid}}$, reinforcing exploitation within high-potential regions:

$$\mathbf{X}_{\text{inew}} = \mathbf{X}_{\text{i}} + rand(0,1)\cdot\left(\mathbf{X}^{\text{best}} - \mathbf{X}_{\text{i}}\right) + rand(0,1)\cdot\left(\mathbf{X}^{\text{best}}_{\text{centroid}} - \mathbf{X}_{\text{i}}\right) \tag{2}$$

When $rand < 0.5$, particles perform a Lévy flight:

$$\mathbf{X}_{\text{inew}} = \mathbf{X}^{\text{best}} + (-1)^{\text{U}(1,2)}\cdot\text{Lévy}(\delta) \tag{3}$$

Lévy flights generate long-tailed step distributions, enabling occasional large displacements across the search space. In the pilgrimage analogy, this corresponds to unpredictable detours of the procession route. From an optimization perspective, it enhances global exploration and mitigates premature convergence to local optima. The parameter δ regulates the heaviness of the step-length distribution and controls the intensity of exploration.

(A) **Leisure ceremony.** This stage refines solutions around promising regions. Instead of perturbing directly around $\mathbf{X}^{\text{best}}$, which may represent a local optimum, particles adjust around $\mathbf{X}^{\text{best}}_{\text{centroid}}$, preserving regional diversity while maintaining exploitation. The mathematical formulation is presented in Eq. (4):

$$\mathbf{X}_{\text{inew}} = \mathbf{X}^{\text{best}}_{\text{centroid}} + (-1)^{\text{U}(1,2)}\cdot rand(0,1)\cdot(\mathbf{Ub} - \mathbf{Lb}) \tag{4}$$

where **Ub** and **Lb** denote the upper and lower search bounds, respectively.

(B) **Crawling beneath the palanquin.** In the pilgrimage ritual, devotees kneel in close succession along the palanquin's path, awaiting its passage overhead. In the algorithm, this behavior is simulated by densely arranging particles around the current best solution, symbolizing collective waiting. This interaction promotes information sharing among particles and strengthens the population's topological connectivity. Consequently, it accelerates convergence and improves the stability of the optimization process. The corresponding mathematical formulation is given in Eq. (5):

$$\mathbf{X}_{\text{inew}} = \mathbf{X}_{\text{i}} + (2\times rand(0,1) - 1)\cdot\left(\mathbf{X}^{\text{best}} - \mathbf{X}_{\text{i}}\right) + (2\times rand(0,1) - 1)\cdot(\mathbf{X}_{\text{r1}} - \mathbf{X}_{\text{i}}) \tag{5}$$

where $\mathbf{X}_{\text{r1}}$ denotes a randomly selected particle from the popu-

lation.

(C) **Palanquin robbing.** In the pilgrimage ritual, this event occurs when a different palanquin team attempts to seize the Mazu palanquin, thereby altering its intended route. In the algorithm, this behavior is represented by a mirrored transformation of particle positions ($\mathbf{X}_i^{rob} = \mathbf{Ub} + \mathbf{Lb} - \mathbf{X}_i$), effectively redistributing them across the search space.

This mechanism increases population diversity and strengthens the algorithm's global search capability, helping it escape local optima and improving overall convergence accuracy. The corresponding mathematical formulation is presented in Eq. (6):

$$\mathbf{X}_{inew} = \mathbf{X}_i^{rob} + rand(0,1)\cdot\left(\mathbf{X}^{best} - \mathbf{X}_i^{rob}\right) + rand(0,1)\cdot\left(\mathbf{X}_{centroid}^{best} - \mathbf{X}_i^{rob}\right) \quad (6)$$

(3) **Return ceremony.** This stage symbolizes the conclusion of the pilgrimage and, in algorithmic terms, represents the final convergence of the search process. At this point, the algorithm identifies and outputs the global optimal solution, completing the optimization cycle.

3.4. Evaluation metrics and procedures

Evaluation addressed two aspects: language model evaluation and instance segmentation evaluation. The MLLM adopted hold-out validation to balance accuracy and computational cost, using training, validation, and test splits with benchmark datasets derived from bridge inspection standards and design references. Semantic quality was evaluated against human references using BERTScore, which computes precision, recall, and F1 scores in the range 0–1 [32], along with multi-class confusion matrices for bridge component and deterioration classification. Instance segmentation performance was assessed using Intersection over Union (IoU), with predictions at IoU greater than or equal to 0.5 treated as true positives, from which Average Precision at IoU 0.5 (AP_{50}) and mean AP_{50} (mAP_{50}) across five classes were computed [33]. Robustness and error analysis were supported by five-fold cross-validation, independent testing [34], and confusion matrices for both YOLO11 and language model outputs [35,36].

4. Model development and evaluation

4.1. Concrete bridge deterioration image dataset

Concrete is the predominant material for bridge structures in Taiwan due to its durability and ease of repair. However, in hot, humid climates, concrete remains vulnerable to deterioration, which can shorten its service life and compromise structural safety. To support this study, image data of deteriorated concrete bridges were compiled from four sources: (1) images previously collected by the research team, (2) annual bridge inspection reports published by the Taipei City Government, (3) the public dataset CODEBRIM (COncrete DEfect BRidge IMage Dataset) [17], and (4) newly captured field images of bridge deterioration.

The integration of these multi-source data ensures the framework's robustness and potential for cross-scenario generalization. Specifically, the inclusion of the international CODEBRIM dataset provides a wide array of concrete surface textures and environmental backgrounds from different geographical contexts. Meanwhile, the official inspection reports from Taipei City encompass bridges with various service lifespans and structural conditions, reflecting real-world complexities. By combining local field data with established international benchmarks, the model is exposed to a diverse feature space, allowing it to adapt to varying concrete material characteristics and aging stages.

Following standard deterioration categories identified in academic literature and practical inspection reports [15], five deterioration types were selected: concrete cracking, concrete spalling, concrete honeycombing, exposed or corroded rebar, and vegetation coverage (Fig. 6**a**). Ten bridge component categories were also included: riverbed, drainage facilities, guardrails, guardrail bases, piers, pier caps, bearing pads, main girders, crossbeams, and bridge decks (Fig. 6**b**).

Although the primary focus is on concrete deterioration, the dataset also includes non-concrete elements, such as riverbeds and drainage systems, as well as environmental factors, including vegetation cover, that influence bridge performance and durability. These components enhance the model's robustness and real-world applicability. The final dataset includes 650 images of deteriorated and 50 images of intact components. The latter were included to train the language model to recognize "non-deteriorated components," enabling the system to distinguish between defective and sound bridge elements.

4.2. Image preprocessing

Bridge images were collected from municipal inspection reports (2022–2023), field surveys, and publicly available datasets. After removing low-quality and non-concrete images, the dataset was categorized into three groups based on content visibility: Type I, identifiable component with deterioration; Type II, deterioration only; and Type III, intact component.

Data annotation followed a dual tool workflow. Text-image pairs were labeled in Label Studio v1.18 using standardized terminology, and pixel-level masks were delineated in LabelMe v5.8.1. The primary annotation was conducted by a graduate student trained in bridge inspection practice. All annotations were reviewed and verified by two faculty members from the civil engineering department, one of whom has over four years of professional experience in bridge engineering and inspection.

To ensure labeling reliability, all annotations were checked against the original images, and ambiguous cases were discussed and resolved through expert review. This two-level verification process was adopted to maintain consistency in defect categorization and boundary delineation. Representative examples of the dataset categories are shown in Figure A1 (Appendix A), and the annotation interfaces for text-image pairs and pixel-level masks are illustrated in Figures A2a and A2b (Appendix A), respectively. Subsequently, custom scripts converted the outputs into JSONL format for fine-tuning Llama 3.2 and into standard YOLO segmentation labels; the complete data preparation pipeline is illustrated in Figure A3 (Appendix A). The relevant source code is provided in Appendices B and C.

4.3. Multimodal language model development and validation

4.3.1. Performance evaluation and metrics

This study employs the Unsloth implementation of the Llama 3.2 11B Vision model as the large language model. Unsloth is an open-source toolkit designed to improve the training efficiency of large-scale models. It enhances fine-tuning performance, reduces memory requirements, and maintains consistency with the original architecture, making it particularly suitable for lightweight fine-tuning tasks. Because fine-tuning in this study was conducted on a single-GPU system (complete source code in Appendix D), Unsloth's efficiency optimizations yielded significant benefits by reducing computational cost and training time.

The model was fine-tuned with 4-bit quantization and the LoRA (Low-Rank Adaptation) approach, reducing the number of trainable parameters and GPU memory requirements. Details on the hardware and software environment are in Appendix E. The core architecture remains aligned with Meta's Llama 3.2 Vision model, with the primary task of generating deterioration descriptions and classifying structural components from bridge images. During the supervised fine-tuning (SFT) stage, the model was trained on a multimodal image–text dataset and evaluated using a hold-out validation set. Training and



a. Deterioration types.

b. Deteriorated bridge components.

Fig. 6. Representative deterioration types and bridge components included in the dataset.

validation loss curves were monitored throughout fine-tuning (Fig. A4, Appendix A).

After fine-tuning, performance was assessed using BERTScore, which compares semantic similarity between generated and reference texts. Evaluations were conducted separately on training and validation sets. To ensure comparability, the maximum number of training steps (max_steps) was set to the same value used later in the PWO optimization experiments. An early-stopping mechanism halted training if the validation loss did not improve for 10 consecutive steps; under this criterion, fine-tuning ended at step 850. Table 1 summarizes the results. Before fine-tuning, the pre-trained model achieved average BERT scores of 0.6025 on the validation set and 0.5929 on the test set. After fine-tuning, the averages increased to 0.8137 and 0.7945, respectively, demonstrating substantial gains in semantic accuracy and generation quality using the proposed dataset.

4.3.2. *Qualitative evaluation of generated sentences*

To supplement the interpretation of quantitative metrics, a qualitative evaluation task was designed to compare differences in the accuracy, clarity, and technical richness of language model outputs. The task used a fixed prompt and an image as input: "Please describe in Traditional Chinese the deteriorated parts and conditions of the bridge based on the image, and explain the repair method." This setup enabled a direct comparison of the Llama 3.2 model before and after fine-tuning, as well as the GPT-4o model, in terms of visual understanding and sentence generation. Table 2 presents the models' outputs for identifying deteriorated parts and types. Both the fine-tuned Llama 3.2 and GPT-4o models successfully identified key defects, such as concrete spalling. In contrast, the non-fine-tuned Llama 3.2 model failed to make correct judgments, producing outputs that lacked meaningful defect recognition.

Table 3 compares repair suggestions generated by the three models. For representative cases, GPT-4o's responses were moderate, while the non-fine-tuned Llama 3.2 produced overly brief sentences lacking technical substance. By contrast, the fine-tuned Llama 3.2 provided detailed conditions and repair recommendations, including technical specifications, indicating improved technical specificity and domain relevance.

Beyond direct comparisons, this study also evaluated outputs along two additional dimensions: language structure and performance stability, as well as engineering knowledge and professional expression (Table 4). Both the fine-tuned Llama 3.2 and GPT-4o demonstrated strong readability through structured bullet points and sentence

Table 1
BERTScore F1 results before and after language model fine-tuning.

	Validation Set		Test Set	
	BERTScore	**Improvement Percentage**	**BERTScore**	**Improvement Percentage**
Before Fine-Tuning	0.6025	-	0.5929	-
After Fine-Tuning	0.8137	+ 35.08%	0.7945	+ 34.01%

Table 2
Deterioration part identification and deterioration-type descriptions generated by Llama 3.2 before and after fine-tuning and by GPT-4o.

Generation Source	**Deterioration Part**	**Deterioration-Type Description**
Content Generated by Llama 3.2 Before Fine-Tuning	NA	NA
Content Generated by Llama 3.2 After Fine-Tuning	Bridge pier	Concrete spalling with exposed and corroded reinforcement.
Content Generated by GPT-4o	Bridge pier	Exposed and corroded reinforcement accompanied by concrete spalling. The concrete cover has been damaged, exposing the reinforcement to air and moisture for an extended period, resulting in severe corrosion that may compromise the member's load-bearing capacity and durability.

Table 3
Repair recommendations generated by Llama 3.2 before and after fine-tuning and by GPT-4o.

Generation Source	Repair Method
Content Generated by Llama 3.2 Before Fine-Tuning	This is corrosion in a reinforced concrete structure, primarily affecting the reinforcement. The concrete surface shows visible cracks, and the internal reinforcement has already corroded, appearing black, indicating severe corrosion of the steel bars.
Content Generated by Llama 3.2 After Fine-Tuning	1. Concrete Removal: Gently chisel the surrounding concrete of the spalled area to fully expose the reinforcement, revealing a clean and stable base concrete. 2. Reinforcement Rust Removal: Use a wire brush, power tools, or sandpaper to clean the corroded reinforcement. 3. Reinforcement Protection: After removing rust, let the surface dry for at least 30 min. Then apply a polymer cement slurry (latex cement slurry) or a zinc-rich coating to prevent re-corrosion, ensuring the coating thickness is within 0.3 mm. 4. Repair Material Preparation: Choose repair materials that have good adhesion and crack resistance, such as polymer-modified mortar or non-shrink cement mortar. 5. Material Application: Fully wet the repair area, apply the repair material in layers, and smooth the surface with a trowel to restore the original appearance of the concrete. 6. Curing: Follow the curing requirements of the repair material, maintaining moisture in the repair area for several days as needed.
Content Generated by GPT-4o	1. Remove Loose or Damaged Concrete: Clear the area down to solid substrate. 2. Reinforcement Rust Removal: Eliminate rust from steel bars through wire brushing, sandblasting, or high-pressure water jetting. 3. Reinforcement Protection: Use anti-corrosion coatings or zinc-based inhibitors to slow down rusting. 4. Reinforcement Repair: When there is a substantial loss in steel cross-section, weld new steel bars or add more ties for reinforcement. 5. Concrete Repair: Rebuild spalled concrete sections with polymer-enhanced or high-adhesion repair mortar. 6. Surface Protection: Use coatings to block water ingress, carbonation, or chloride penetration. 7. Structural Strengthening: For extensive damage or insufficient structural performance, consider external steel plates, carbon fiber reinforcement, or other structural strengthening methods.

Table 4
Language structure, performance stability, and engineering knowledge expression between GPT-4o and the fine-tuned Llama 3.2 model.

Comparison Item	GPT-4o	Fine-Tuned Llama 3.2	Outcome
Clarity and Structure of Bullet Points	Clearly itemized with a clear structure	Fully itemized with well-segmented sentences	Comparable performance
Semantic Fluency and Grammatical Accuracy	Sentences are smooth, grammatically consistent, with almost no redundant words	Sentences are smooth, but occasionally contain redundancies or typos	GPT-4o superior
Response Consistency	Response style is stable with high consistency	Generated sentences vary significantly between each response	GPT-4o superior
Ability to Express Technical Parameter Details	Most responses lack specific numerical values, only generally describing methods or material types	Clearly specifies technical details such as 30-minute drying time and 0.3 mm coating thickness	Llama 3.2 superior
Degree of Explanation of Material Application Methods	Although material types are mentioned, specific application procedures or control methods are not explained	Clearly explains the application process, such as coating sequence, thickness control, and curing recommendations	Llama 3.2 superior
Professionalism and Logical Consistency of Semantics	Formal tone with clear narrative logic	Formal tone, aligned with engineering terminology	Comparable performance

Table 5
Category-wise comparison of YOLO11-seg mAP_{50} performance on the validation and test datasets.

Category	Validation Set		Test Set	
	mAP_{50} (%)	Rank	mAP_{50} (%)	Rank
Overall Average	50.5	-	52.1	-
Concrete Honeycombing	50.8	2	49.3	3
Concrete Spalling	50.3	3	48.4	4
Concrete Cracking	38.0	5	55.6	2
Exposed & Rusted Rebar	42.0	4	36.3	5
Vegetation Coverage	71.4	1	71.0	1

hierarchy, though GPT-4o showed greater grammatical stability and consistency. From a technical standpoint, the fine-tuned Llama 3.2 produced content enriched with numerical values and references to materials, with a tone closer to engineering writing norms. Overall, its fine-tuned outputs were rated as more aligned with engineering-style reporting in terms of professionalism and practicality.

4.4. Instance segmentation model development and validation

This study employed the YOLO11-seg model to develop and train an instance segmentation framework for concrete bridge deterioration (complete implementation is provided in Appendix F). The model was trained to detect five types of deterioration: concrete honeycombing, concrete spalling, concrete cracking, exposed or corroded reinforcement, and vegetation coverage. To ensure stability and generalization, performance was evaluated using a five-fold cross-validation procedure across both validation and independent test sets.

Table 5 presents the mean Average Precision at IoU 0.5 (mAP_{50}) for each defect category. On average, the model achieved an mAP_{50} of 50.5% on the validation set and 52.1% on the test set, demonstrating a reliable ability to identify diverse defect types. The highest accuracy was obtained for vegetation coverage, likely due to its distinctive texture and clear visual boundaries. Detection of concrete honeycombing and spalling also yielded consistent results across folds.

By contrast, performance on concrete cracking and exposed reinforcement showed greater variability. These categories exhibit fine textures and irregular patterns that are easily obscured by surrounding structural elements, thereby reducing segmentation accuracy. Although the overall mAP_{50} is reasonable, the reported performance is consistent with the challenging nature of concrete defect detection tasks and falls within the range reported in recent studies [16]. These findings highlight the inherent difficulty of identifying subtle concrete-related deterioration in computer vision tasks. Future improvements are expected through refined annotations, larger datasets, and continued model optimization.

4.5. Metaheuristic hyperparameter optimization

In this study, the PWO algorithm was applied to fine-tune the

hyperparameters of the Llama 3.2 model, focusing on five key parameters (see the full script in Appendix D). The objective was to improve the quality of bridge-related image descriptions as measured by the BERTScore validation metric. PWO was selected as a gradient-free metaheuristic capable of exploring nonconvex, moderately high-dimensional hyperparameter spaces without gradients. This property is particularly suitable to fine-tuning large language models, where convergence behavior and semantic alignment are sensitive to learning-rate scheduling and regularization settings. Under a fixed evaluation budget, PWO automated hyperparameter selection, reducing manual trial-and-error.

Among the tuned parameters, the maximum training steps (max_steps) lacked a predefined default value; therefore, the baseline experiment used the same value as in the PWO optimization for consistency. The hyperparameter ranges, along with their initial and optimized values, are summarized in Table A2 (Appendix A). The entire optimization process required approximately 180 h, during which PWO identified the best-performing hyperparameter combination. This optimized configuration was subsequently used for fine-tuning the model.

To mitigate overfitting, an early-stopping mechanism was employed, consistent with the baseline setup. Training was halted if the validation loss failed to improve for 10 consecutive evaluations. Under this criterion, the model stopped at step 850. As illustrated in Figure A5 (Appendix A), comparisons of training and validation loss curves before and after optimization demonstrate that the PWO-derived parameters indicate improved training stability after tuning under the same early-stopping criterion.

As shown in Table 6, although both PWO and GA searched within similar predefined ranges, the optimized hyperparameters differed noticeably, reflecting distinct search behaviors under the same evaluation budget. The search ranges were defined based on the default and recommended LoRA fine-tuning configurations provided in the Unsloth documentation for large language models and were moderately expanded to ensure sufficient exploration of the hyperparameter space. It should be noted that max_steps represents the optimal upper bound found, while the actual training process continues to be governed by the same early-stopping criterion. As shown in Table 7, both GA and PWO further improved BERTScore over the baseline after fine-tuning on the dataset. The consistent gains on validation and test sets indicate improved generalization performance. Among them, PWO achieved the highest improvement percentages, suggesting stronger effectiveness in this hyperparameter optimization setting.

As in the language model optimization procedure, the PWO algorithm was used to tune the hyperparameters of the YOLO11-seg model. A total of 23 hyperparameters were optimized, covering learning rate scheduling, loss-related coefficients, and data augmentation parameters (Appendix G). All hyperparameters were initialized using the official YOLO11 default settings, and the predefined search ranges were set accordingly to ensure consistency with the original training configuration.

Attention was given to augmentation-related parameters, including HSV gains, geometric transformations, and mosaic and mixup strategies. Data augmentation was treated as an optimizable component of training rather than an arbitrary image-expansion strategy. This design was motivated by the authors' previous bridge deterioration annotation study [37], in which PWO Lite was used to optimize YOLOv8 augmentation parameters for UAV-captured images of bridge deck corrosion. The optimized results retained moderate HSV adjustments, translation, scaling, horizontal flipping, and mosaic augmentation, while suppressing rotation, shear, perspective transformation, vertical flipping, BGR conversion, mixup, and copy-paste. These findings indicate that excessive or physically unrealistic augmentation may generate image conditions rarely encountered in routine bridge inspections and may reduce engineering relevance.

Therefore, in the present study, the augmentation-related parameters of YOLO11-seg were optimized using PWO within predefined ranges rather than being manually imposed. These augmentations encourage the model to learn invariant representations of deterioration features under varying illumination, shadows, and occlusions. After approximately 130 h of iterative search, PWO identified the best-performing configuration. The predefined search ranges and the optimized results obtained using GA and PWO are summarized in Table 8. The differences between GA-optimized and PWO-optimized values reflect distinct search trajectories within the same parameter space.

As shown in Table 9, hyperparameter tuning consistently improved segmentation accuracy. The overall average mAP_{50} increased from 50.5% to 53.6% on the validation set and from 52.1% to 54.0% on the test set after PWO optimization. GA tuning also improved performance, though the gains were comparatively smaller. Category-level analysis indicates that the most notable improvements occurred in concrete cracking and vegetation coverage. For instance, cracking performance increased from 38.0% to 41.9% on the validation set and from 55.6% to 58.1% on the test set after PWO tuning. These results suggest enhanced sensitivity to both fine-scale defects and visually distinctive surface conditions. The metaheuristic-based optimization improved generalization capability and detection stability across categories. The performance gain should therefore be interpreted as the effect of optimized augmentation-related training and hyperparameter tuning, rather than as evidence that synthetic augmentation can fully resolve the limitations of a limited dataset.

Although the same PWO framework was applied to both models, their optimization behaviors differed due to intrinsic differences between generative language modeling and instance-level segmentation. For Llama 3.2, the tuned hyperparameters primarily influenced convergence dynamics and semantic alignment within a relatively low-dimensional search space. In contrast, YOLO11-seg involved a higher-dimensional search space that included loss-balancing and augmentation parameters, which interacted nonlinearly with spatial feature extraction. Therefore, the observed disparities in optimized parameter distributions reflect differences in task objectives, loss functions, and optimization landscapes rather than the PWO mechanism itself. This

Table 6
Hyperparameter search ranges and optimized values for the Llama 3.2 Vision model.

Hyperparameter	Range	PWO-Optimized Value	GA-Optimized Value
learning_rate	[1e-5, 1e-3]	5.11e-4	1.17e-4
max_grad_norm	[0.1, 1.0]	0.498	0.826
weight_decay	[0.00, 0.10]	0.0504	0.0017
max_steps	PWO: [400, 2000]; GA: [300, 2000]	1390	327
warmup_ratio	[0.01, 0.10]	0.0879	0.0222

Table 7
BERTScore evaluation results before and after hyperparameter optimization.

	Validation Set		Test Set	
	BERTScore	Improvement Percentage	BERTScore	Improvement Percentage
Baseline Fine-Tuned Model	0.8137	-	0.7945	-
GA-Optimized Fine-Tuned Model	0.8362	+ 2.77%	0.8228	+ 3.56%
PWO-Optimized Fine-Tuned Model	0.8523	+ 4.74%	0.8384	+ 5.53%

Table 8
YOLO11-seg hyperparameter search ranges and optimized values.

Hyperparameter	Range	Original Setting	GA-Optimized Value	PWO-Optimized Value
lr0	[3.16e-4, 2e-2]	0.01	0.0035	0.0076
lrf	[0.01, 0.3]	0.01	0.1274	0.0603
momentum	[0.85, 0.97]	0.937	0.8805	0.899
weight_decay	[3.16e-5, 1e-3]	0.0005	0.0005	0.0002
warmup_epochs (epochs)	[0.0, 5.0]	3.0	2.1735	3.1338
warmup_momentum	[0.7, 0.9]	0.8	0.7978	0.8653
box	[5.0, 10.0]	7.5	6.1828	7.5897
cls	[0.3, 1.2]	0.5	1.1146	0.8662
dfl	[1.0, 2.0]	1.5	1.2042	1.1410
hsv_h	[0.0, 0.05]	0.015	0.0252	0.0358
hsv_s	[0.4, 0.9]	0.7	0.5409	0.6158
hsv_v	[0.2, 0.7]	0.4	0.4729	0.3504
degrees	[0.0, 20.0]	0.0	3.0832	14.9510
translate	[0.0, 0.3]	0.1	0.1595	0.1430
scale	[0.4, 0.8]	0.5	0.6306	0.5731
shear	[0.0, 5.0]	0.0	1.7683	2.1664
perspective	[0.0, 0.005]	0.0	0.0018	0.0
flipud	[0.0, 0.3]	0.0	0.2847	0.0552
fliplr	[0.3, 1.0]	0.5	0.9941	0.4737
mosaic	[0.8, 1.0]	1.0	0.8989	0.8733
mixup	[0.0, 0.3]	0.0	0.0534	0.1526
copy_paste	[0.0, 0.3]	0.0	0.0455	0.0537
bgr	[0.0, 0.5]	0.0	0.3106	0.1857

suggests that the metaheuristic search adapts to model-specific characteristics under the same optimization strategy.

4.6. Vision–language integration mechanism

The proposed framework integrates an instance segmentation model and a multimodal language model via a structured decision-level synchronization mechanism. For each input image, the YOLO11-seg model first performs instance-level detection and then generates pixel-level masks. Post-processing aggregates mask areas by defect category and converts pixel counts into quantitative indicators, including deterioration area ratios and estimated repair costs. These numerical outputs are serialized into predefined statistical fields.

In parallel, the same image is processed by the fine-tuned Llama 3.2 Vision model to generate structured textual outputs describing the bridge component, deterioration status, and recommended repair measures. A template-based parsing scheme is applied to extract component identifiers and intervention descriptions from the generated text. The segmentation-derived quantitative indicators and the language model outputs are then stored within a unified data structure, ensuring consistent association between visual measurements and narrative descriptions.

To maintain cross-modal consistency, synchronization is performed at the record level rather than at the feature-embedding level. All outputs follow a fixed schema that includes component category, deterioration type, quantitative area, and cost estimation. When multiple defect instances are detected in the same image, they are aggregated by category before being injected into the reporting template to avoid prompt overload and inconsistent mapping. This aggregation strategy ensures that numerical values remain aligned with their corresponding semantic descriptions.

Potential synchronization challenges primarily involve semantic alignment and structured consistency between pixel-level measurements and language-level abstraction. These are mitigated through standardized label taxonomies, fixed output fields, and confidence-based aggregation. By adopting a decision-level integration strategy, the framework achieves coherent multimodal reporting without requiring joint end-to-end training across vision and language modules.

4.7. Comparative performance analysis

For instance segmentation, the optimized YOLO11-seg model achieved an overall mAP_{50} of 53.6% on the validation set and 54.0% on the test set after PWO tuning. Recent bridge crack instance segmentation studies reported substantially higher segmentation mAP_{50} on crack-specific datasets. For example, Assad et al. reported segmentation mAP_{50} of 87.8% for YOLOv9 and 87.1% for YOLO11, while Mask R-CNN achieved 75.5% under the same experimental setting [38]. These results are typically obtained on focused crack datasets and are often accompanied by deployment-oriented reporting, such as inference time in milliseconds per image. In contrast, the present study targets five deterioration categories across heterogeneous bridge components and backgrounds, in which fine-texture defects and ambiguous boundaries are common sources of error. The dacl10k benchmark further illustrates the intrinsic difficulty of multi-defect bridge-inspection segmentation, reporting mean IoU values of approximately 0.414 on the validation split and 0.424 on the test split for baseline semantic segmentation models [39]. These benchmarks provide a consistent reference: multi-class bridge damage segmentation is substantially more challenging than single-defect crack segmentation.

For the multimodal language stream, dataset fine-tuning improved BERTScore from 0.603 to 0.814 on the validation set and from 0.593 to 0.795 on the test set, and further improved to 0.852 and 0.838 after PWO optimization. Prior work on bridge damage captioning and defect

Table 9
YOLO11-seg evaluation metrics before and after hyperparameter optimization.

Category	Validation Set			Test Set		
	mAP_{50} (%)	mAP_{50} after GA (%)	mAP_{50} after PWO (%)	mAP_{50} (%)	mAP_{50} after GA (%)	mAP_{50} after PWO (%)
Overall Average	50.5	52.7	53.6	52.1	53.3	54.0
Concrete Honeycombing	50.8	50.9	50.6	49.3	51.3	52.4
Concrete Spalling	50.3	53.1	55.2	48.4	46.9	47.4
Concrete Cracking	38.0	40.5	41.9	55.6	57.0	58.1
Exposed & Rusted Rebar	42.0	44.0	44.9	36.3	37.6	38.3
Vegetation Coverage	71.4	75.0	75.2	71.0	73.7	73.8

description generation commonly reports lexical overlap metrics such as BLEU, METEOR, ROUGE-L, and CIDEr. For example, Chun et al. reported BLEU scores of 0.782, 0.749, 0.711, and 0.693 for bridge damage captioning, indicating strong n-gram level agreement on their dataset [40]. Similarly, Chai et al. reported BLEU-1 of 0.703, BLEU-4 of 0.575, METEOR of 0.399, ROUGE-L of 0.722, and CIDEr of 5.583 for bridge defect description generation with an attention-based encoder-decoder model [41]. In this context, the proposed framework uses BERTScore to emphasize semantic similarity in technical reporting. This metric is complemented by a questionnaire-based comparison against a general-purpose model under identical prompts, providing both objective and user-centered evidence of improved engineering consistency.

Compared with prior studies that typically treat segmentation and defect narration as separate tasks, the proposed framework integrates multi-defect instance segmentation, quantitative measurement, and retrieval-augmented narrative generation into a single inspection-oriented workflow. The combined results support stable, multi-category deterioration documentation and traceable reporting suitable for practical bridge maintenance decision-making.

5. Development of the inspection report generation system

5.1. Real-world case study of inspection report generation

To demonstrate the practical applicability of the proposed framework, a representative field inspection case is presented. The input image shows a bridge pier with visible concrete spalling and localized exposed reinforcement, collected during a routine inspection. Deterioration is first localized using YOLO11-seg, which generates polygon masks with normalized coordinates (x_i, y_i). These are scaled by image width and height (*Wpx, Hpx*) to obtain pixel coordinates (X_i, Y_i), and the pixel area ($area_{px}$) is calculated using the *shapely.geometry.Polygon* function. A default magnification factor of 1 is assumed, with optional calibration enabled through a known scale within the image to derive the pixel-to-real-world conversion factor s based on lens and ruler geometry (Eq. 7), where d and h are real-world reference dimensions, and D and H are the corresponding pixel dimensions in the image. The actual defect area is then calculated as $area_{px} \times s^2$ (Eq. 8). Areas are multiplied by user-editable unit prices *Pclass* (NT$/m^2) to estimate repair costs, which are displayed alongside per-defect areas in the system output.

$$s = \frac{d}{D} = \frac{h}{H} \tag{7}$$

$$area_{m^2} = area_{px} \times s^2 \tag{8}$$

The interface was developed as a Gradio web application, with complete source code provided in Appendix H. The workflow integrates bridge metadata input, cost parameter configuration, instance segmentation, quantitative area computation, retrieval-augmented language generation, and PDF export into a unified reporting process. The RAG module references the Ministry of Transportation Specifications to provide standardized repair guidance. All runtime measurements were conducted on an NVIDIA RTX A6000 GPU under identical hardware conditions. Under this configuration, the end-to-end processing time ranges from 30–50 s per image. This duration includes segmentation inference, quantitative computation, retrieval, language generation, and the synthesis of structured reports. The complete interface and operational workflow are illustrated in Fig. 7.

Segmentation-only pipelines on modern GPUs typically achieve inference times ranging from tens to hundreds of milliseconds per image, as they primarily involve forward-pass detection. In contrast, the proposed framework performs additional numerical analysis, retrieval-augmented reasoning, and document generation within a unified execution workflow. The increased processing time is therefore primarily attributable to multimodal integration rather than detection latency alone.

Further acceleration remains feasible through mixed precision inference, model compression, parallel execution of independent modules, and batch-based processing strategies. In practical bridge inspection scenarios, where images are processed individually rather than in real-time streams, the current latency remains operationally acceptable. A representative automatically generated inspection report is shown in Fig. 8, demonstrating the integration of segmented imagery, deterioration descriptions, and cost summaries within a standardized reporting format.

5.2. Vision-to-language tokenization and prompt engineering

To bridge the gap between visual segmentation outputs and structured engineering reports, a structured data conversion process transforms raw pixel-level measurements into linguistic tokens that serve as conditioning inputs for the language model. The framework extracts key attributes from the YOLO11-seg module, including structural component labels (e.g., pier, girder, deck), defect classifications (e.g., cracking, spalling), and quantitative metrics. These metrics are derived from pixel-level masks to compute deterioration area and corresponding repair cost estimates, ensuring that qualitative visual findings are grounded in quantitative engineering measurements.

The extracted attributes are serialized into a standardized schema and embedded within a domain-specific prompt template. The template provides explicit contextual guidance to the fine-tuned Llama 3.2 Vision model, for example: “As a certified bridge inspector, analyze the detected [Defect Type] on the [Component] with a measured area of [Area] m^2. Based on the relevant engineering guidelines [RAG context], describe the deterioration and recommend prioritized repair actions.” This structured conditioning mechanism ensures that the model receives precise, domain-relevant input rather than unstructured visual data, thereby improving the consistency and technical coherence of generated reports.

5.3. User evaluation and system feedback

To validate the practical performance of the fine-tuned Llama 3.2 Vision model in bridge defect description and repair suggestion tasks, this study conducted a questionnaire-based comparative evaluation using GPT-4o as the benchmark. The analysis therefore compared Llama 3.2 Vision and GPT-4o to assess whether the fine-tuned model can provide usable, engineering-consistent narratives, using GPT-4o as the reference baseline under the same prompt settings.

The questionnaire was designed around outputs generated from bridge deterioration images. It included four evaluation criteria, each rated on a five-point Likert scale: (1) accuracy of defect component identification and scenario description, (2) professionalism and feasibility of repair suggestions, (3) technical clarity and precision of language, and (4) overall completeness and applicability (Appendix I). Respondents, all with backgrounds in civil engineering or construction, performed blind evaluations to minimize bias. Results were analyzed using average score comparisons and paired-sample t-tests.

A total of 16 valid responses were collected. Participants' professional backgrounds were distributed as follows: 37.5% office-based, 31.2% combined office and field, 18.7% field-only, and 12.5% with no practical experience. As shown in Table 10, the fine-tuned Llama 3.2 Vision model achieved consistently higher mean ratings than GPT-4o across all four evaluation criteria. The largest improvements were in technical clarity and language precision (4.12 vs. 3.61) and overall content completeness and applicability (4.08 vs. 3.61). Paired-sample t-tests confirmed that the two differences were statistically significant ($p = 0.003$ and $p = 0.011$, respectively), and that the improvement in repair-method professionalism was also significant ($p = 0.039$). In contrast, the difference in defect identification accuracy did not reach statistical significance ($p = 0.281$), suggesting that both models perform comparably in basic visual recognition but differ notably in domain-

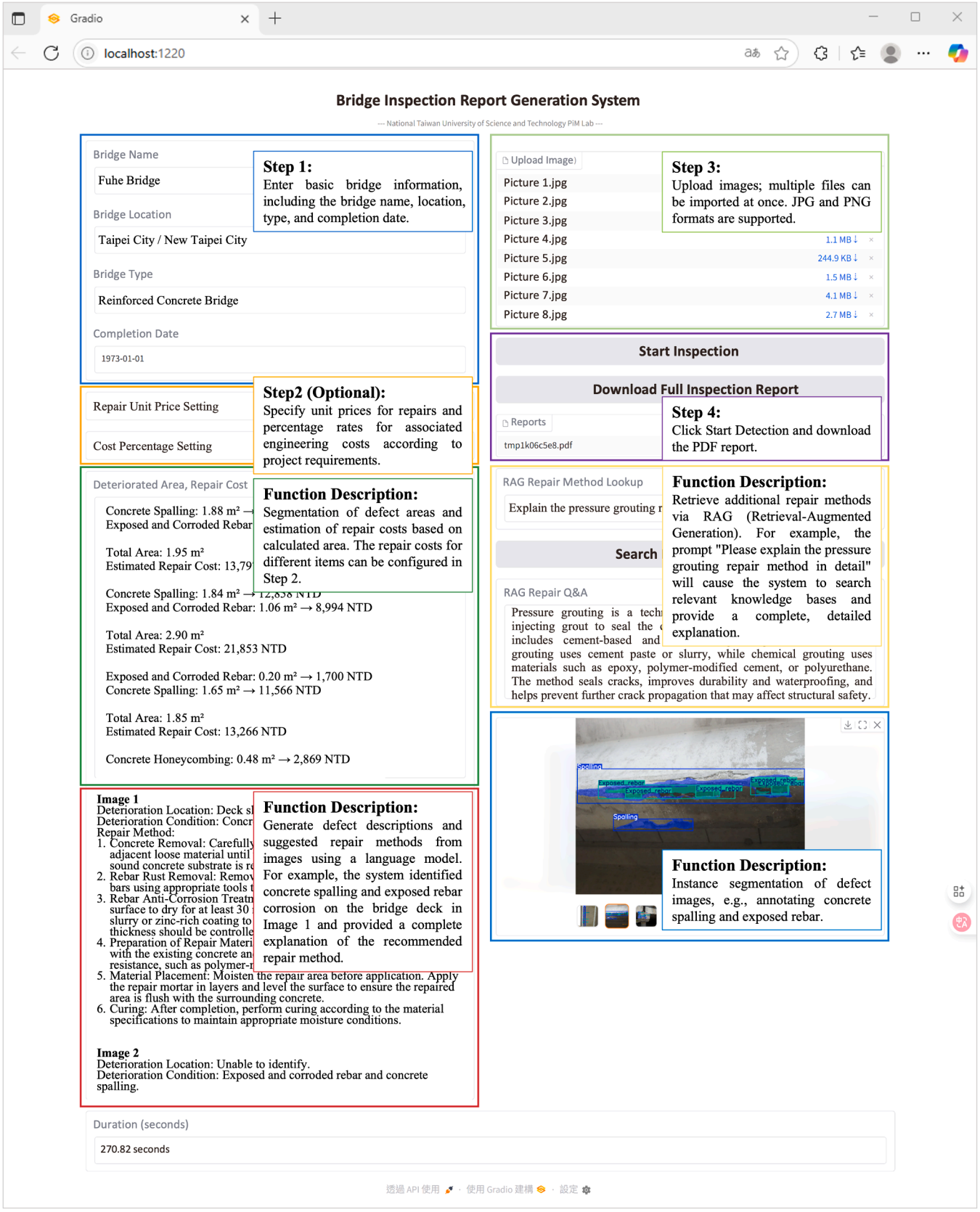


Fig. 7. User interface of the automated concrete bridge inspection report generation system.

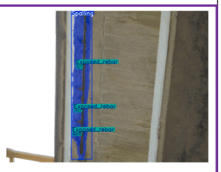
Bridge Inspection Report

Basic Bridge Information, including the bridge name, location, type, completion date, and inspection date.

Bridge Name	Fuhe Bridge
Bridge Location	Taipei City / New Taipei City
Bridge Type	Reinforced Concrete Bridge
Completion Date	1973-01-01
Inspection Date	2025-07-01

Common inspection report items: component, result, repair method, area, cost, and image.

Detailed inspection report content

Item	Component	Result	Recommended Repair Method	Area / Cost	Image
I	Contracted Project Cost				
1	Bridge Project			34,715 NTD	
1.1	Deck Slab	Concrete spalling and exposed, corroded rebar	1. Concrete Removal: Carefully remove spalled and loose concrete until the reinforcing bars are fully exposed and sound concrete substrate is reached. 2. Rebar Rust Removal: Remove rust from the reinforcing bars using appropriate tools to ensure a clean surface free of corrosion. 3. Rebar Protection: After rust removal, allow the surface to dry for at least 30 minutes, then apply polymer cement slurry or zinc-rich coating to prevent further corrosion. The coating thickness should be controlled within 0.3 mm. 4. Repair Material Preparation: Select repair materials compatible with the original concrete and possessing good adhesion and crack resistance, such as polymer-modified mortar or non-shrink mortar. 5. Material Application: Moisten the repair area before application. Apply repair mortar in layers and level the surface to ensure it is flush with the surrounding concrete. 6. Curing: After completion, perform curing according to material specifications to maintain proper moisture conditions.	Concrete spalling: 1.88 m² → 13,149 NTD Exposed and corroded rebar: 0.08 m² → 647 NTD Total area: 1.96 m² Estimated repair cost: 13,796 NTD	
1.2	Cross Beam	Concrete honeycombing	1. Define repair area and remove unsound concrete: Chisel away deteriorated concrete until sound substrate is reached. The removal surface should be roughened. If exposed aggregate is present, the removal depth should not exceed 1 cm, or the surface slope should not be less than 1:1. 2. Surface preparation: Roughen the repair area. If cracks are present, extend the repair area at least 5 cm beyond the crack, with cutting depth not exceeding 1 cm. 3. Surface cleaning: Use high-pressure water or wire brushing to remove dust and loose particles, ensuring the surface is free from oil and debris. 4. Bonding agent application: After drying, apply bonding agent to enhance adhesion between repair material and existing concrete. Common materials include epoxy resin or cement slurry. 5. Selection of repair materials: Choose high-strength, low-shrinkage, durable repair materials, such as non-shrink cement mortar or resin mortar. 6. Material placement: Before the bonding agent hardens, apply repair material to the prepared area and finish to level. 7. Curing: Apply moist curing or protective covering to prevent rapid moisture loss and cracking.	Concrete honeycombing: 3.49 m² → 20,919 NTD Total area: 3.49 m² Estimated repair cost: 20,919 NTD	
1.3	Bridge Pier, Bearings, and Main Girders	No deterioration detected in the structural components shown in the image	No repair required.	(No deterioration detected) Total area: 0.00 m² Estimated repair cost: 0 NTD	
2	Miscellaneous Works (25% of item 1)			8,679 NTD	
3	Labor Safety and Health Equipment (25% of item 2)			2,170 NTD	
II	Other Project Costs				
1	Air Pollution Control Fee (8% of Contracted Project Cost)			3,645 NTD	
2	Project Management Fee (10% of Contracted Project Cost)			4,556 NTD	
3	Temporary Electricity Fee (8% of Contracted Project Cost)			3,645 NTD	
	Total (Contracted Project Cost and Other Project Costs)			57,410 NTD	

Automatically calculate additional costs based on repair costs in each image, including miscellaneous works, labor safety equipment, air pollution control, project management, temporary electricity, and total cost.

Fig. 8. Automatically generated inspection report for concrete bridge deterioration.

Table 10
Evaluation scores for generated content produced by the Llama 3.2 Vision and GPT-4o models.

Evaluation Criteria	Model A (Llama 3.2 Vision) Mean / Std. Dev.	Model B (GPT-4o) Mean / Std. Dev.	Paired Sample t-Test (p − value)
Accuracy of Defect Component Identification and Defect Scenario Description	3.92 / 0.49	3.71 / 0.61	0.281
Professionalism and Practical Feasibility of the Suggested Repair Methods	4.03 / 0.52	3.70 / 0.56	0.039
Technical Clarity and Precision of Language Used	4.12 / 0.51	3.61 / 0.60	0.003
Overall Content Completeness and Applicability	4.08 / 0.45	3.61 / 0.57	0.011
Overall Category Average	**4.037**	**3.658**	**0.0165**

Table 11
Statistical results of the system user experience evaluation.

Evaluation Criteria	Average Score	Standard Deviation
The system interface is straightforward to understand	4.313	0.704
The operation process is intuitive and smooth	4.250	0.447
Overall system execution is stable and smooth	4.375	0.619
Image recognition and report generation speeds are satisfactory	4.125	0.619
Overall experience is positive, and the system would be recommended to others	4.375	0.719

specific language generation and engineering applicability.

For the accuracy of defect component identification and scenario description, no significant difference was found ($p = 0.281$). Respondent comments suggested that while GPT-4o showed slightly stronger performance in component identification, Llama 3.2 Vision excelled in providing detailed scenario descriptions. This divergence in strengths likely contributed to the variation in scores. Overall, Llama 3.2 Vision achieved an average score of 4.04, outperforming GPT-4o's 3.66, with statistically significant advantages in key dimensions of linguistic expression and structural coherence.

To further assess usability, the questionnaire included five additional items on system design and user experience, rated on a five-point Likert scale (Table 11). Results indicated strong user approval, with the highest ratings for "system operation is stable and smooth" and "overall experience is good, and I would recommend its use," both averaging 4.375. The lowest score, "image recognition and report generation speed is satisfactory," received an average of 4.125, reflecting user sensitivity to longer inference times. Nevertheless, this still represents an upper-middle performance level. Overall, participants rated both the system's operational stability and practical usability highly.

6. Conclusion

This study presented a non-destructive evaluation framework that automates the detection, quantification, and documentation of concrete bridge deterioration by integrating a YOLO11-seg instance segmentation model with a Llama 3.2 Vision language model fine-tuned via Low-Rank Adaptation (LoRA), with both modules enhanced through the Pilgrimage Walk Optimization (PWO) algorithm. The system links pixel-level defect masks to quantitative deterioration area estimates and corresponding repair costs. At the same time, the Retrieval-Augmented Generation (RAG)–enhanced language model generates standardized descriptions and repair recommendations consistent with engineering guidelines. Implemented locally via a browser interface, the framework produces structured, traceable inspection outputs suitable for practical bridge assessment.

Across the evaluated dataset, LoRA-based fine-tuning markedly improved the language model's descriptive accuracy, with BERTScore increases of 35.08% on the validation set and 34.01% on the test set (Table 1), followed by an additional improvement of 4.74%–5.53% after PWO optimization (Table 7). For instance segmentation, PWO improved mAP_{50} by 6.14% on validation and 3.65% on testing relative to the baseline (Table 9). Under identical prompts, the fine-tuned model generated defect descriptions and repair suggestions comparable to those of GPT-4o, although differences in linguistic structure and technical detail remained. These results indicate that metaheuristic enhancement strengthens defect quantification and increases the reliability of deterioration-related documentation.

Operationally, the framework consolidates image acquisition, defect detection, deterioration quantification, and narrative generation into a reproducible non-destructive evaluation pipeline for concrete bridges. Adjustable unit costs, configurable templates, and traceable outputs support consistent repair estimation and facilitate comparison across inspection cycles. Its modular design also allows detection models, language components, cost parameters, and retrieval sources to be updated as deterioration datasets and engineering standards evolve.

Despite these promising results, several limitations should be acknowledged. First, the dataset remains moderate, comprising 700 images in total, which may limit its ability to represent rare deterioration patterns and complex structural configurations. Although the dataset integrates multiple sources, including municipal inspection records and publicly available benchmarks, most images originate from bridge inspections in Taiwan. As a result, the data distribution reflects predominantly humid, typhoon-prone climatic conditions and reinforced-concrete structures typical of this regional context.

Second, the model's generalization capability across diverse environmental and structural conditions has not yet been systematically validated. Variations in material properties, freeze–thaw cycles, deicing salt exposure, and inspection imaging protocols may affect both segmentation accuracy and the consistency of language generation. Although data augmentation was used to simulate variations in lighting, occlusion, and background interference, these transformations cannot fully replace geographically diverse real-world datasets. Future validation should therefore incorporate independent datasets from different climatic regions and bridge systems to assess transferability, robustness, and potential performance degradation under unseen conditions.

Third, the current cost estimation approach relies on user-defined unit prices and area-based calculations. While this design ensures transparency and traceability, it does not account for historical maintenance data or predictive cost modeling that reflects market variability and structural condition indices. Integrating large-scale infrastructure maintenance records and data-driven cost models could further strengthen the system's economic decision-support capability.

Future work will focus on expanding the dataset to include a broader range of climatic regions, deterioration types, and structural systems, and on conducting cross-regional validation. In addition, incorporating joint vision–language fine-tuning, uncertainty quantification, and data-driven cost prediction may enhance robustness and support more reliable engineering decision-making in practical bridge assessment applications.

Replication of results

The datasets, source code, and replication results generated and analyzed in this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Jui-Sheng Chou: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources,

Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Chia-Hsiang Chang:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Chi-Yun Liu:** Writing – review & editing, Visualization, Validation, Software, Methodology, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.engstruct.2026.122898.

Data Availability

Data will be made available on request.

References

[1] Ministry of Labor, 2024 Human Resources Demand Survey (Third Round), 2024. ⟨https://statdb.mol.gov.tw/html/s3/svy13/1333menu.htm⟩ (accessed March 1, 2025).
[2] McKinsey Global Institute, Reinventing Construction: A Route to Higher Productivity, 2017. Available: ⟨https://www.mckinsey.com/capabilities/operations/our-insights/reinventing-construction-through-a-productivity-revolution⟩.
[3] Institute of Transportation, Ministry of Transportation and Communications, National Bridge Classification Statistics. ⟨https://bss.iot.gov.tw/bss/country/statistics_all/list/⟩ (accessed March 1, 2025).
[4] M. Alsharqawi, T. Zayed, S. Abu Dabous, Common Practices in Assessing Conditions of Concrete Bridges, *MATEC Web of Conferences*, vol. 120, p. 02016, 2017, doi: 10.1051/matecconf/201712002016.
[5] Abdallah AM, Atadero RA, Ozbek ME. A State-of-the-Art Review of Bridge Inspection Planning: Current Situation and Future Needs. J Bridge Eng 2022;27(2):03121001. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001812.
[6] Chan CF, Wong PK-Y, Guo X, Cheng JCP, Chan JP-C, Leung P-H, Tao X. Context-aware vision-language model agent enriched with domain-specific ontology for construction site safety monitoring. Autom Constr 2025;177:106305. https://doi.org/10.1016/j.autcon.2025.106305.
[7] Kim T, Kim S, Chern W-C, Park S, Kim D, Kim H. Optimizing large vision-language models for context-aware construction safety assessment. Autom Constr 2025;180:106510. https://doi.org/10.1016/j.autcon.2025.106510.
[8] Lee J, Ahn S, Kim D, Kim D. Performance comparison of retrieval-augmented generation and fine-tuned large language models for construction safety management knowledge retrieval. Autom Constr 2024;168:105846. https://doi.org/10.1016/j.autcon.2024.105846.
[9] M. Moore, B.M. Phares, B. Graybeal, D. Rolander, G. Washer, Reliability of Visual Inspection for Highway Bridges, Volume I: Final Report, Rep. FHWA-RD-01-020, Federal Highway Administration, McLean, VA, 2001. Available: ⟨https://rosap.ntl.bts.gov/view/dot/33883⟩.
[10] Hamida Z, Goulet J-A. Quantifying the relative change in maintenance costs due to delayed maintenance actions in transportation infrastructure. J Perform Constr Facil 2024;38(5):04024035. https://doi.org/10.1061/JPCFEV.CFENG-4802.
[11] Jung Y, Cho I, Hsu S-H, Golparvar-Fard M. VisualSiteDiary: a detector-free Vision-Language Transformer model for captioning photologs for daily construction reporting and image retrievals. Autom Constr 2024;165:105483. https://doi.org/10.1016/j.autcon.2024.105483.
[12] Li T, Wang H, Tan J, Kong L, Zhang H, Pan D, Zhao Z. Intelligent quality assessment of concrete vibration using computer vision and large language models. Autom Constr 2025;180:106507. https://doi.org/10.1016/j.autcon.2025.106507.
[13] Institute of Transportation, Ministry of Transportation and Communications, Second Generation Taiwan Area Bridge Management Information System Construction Planning (Part III), Rep. no. 107-033-1398, Institute of Transportation, Taipei, Taiwan, 2018. Available: ⟨https://www.iot.gov.tw/uploads/asset/data/661975e3367376304acd4e01/%E7%AC%AC%E4%BA%8C%E4%BB%A3%E8%87%BA%E7%81%A3%E5%9C%B0%E5%8D%80%E6%A9%8B%E6%A2%81%E7%AE%A1%E7%90%86%E8%B3%87%E8%A8%8A%E7%B3%BB%E7%B5%B1%E5%BB%BA%E7%BD%AE%E8%A6%8F%E5%8A%83(%E4%B8%89).pdf⟩.
[14] Maintenance and Upkeep Section, New Construction Office, Public Works Department, Taipei City Government, 111–112 Bridge Key Inspection Results (Maintenance Section), Taipei City Government, Taipei, Taiwan, 2023. Available: ⟨https://nco.gov.taipei/News.aspx?n=E150D8278BFA8D52&sms=79B5F59149C69A28⟩.
[15] Tsai H.-C., Wang H.-S., Mao I.-S., Yeh C.-H., Wang C.-H. Discussion on common deterioration types of concrete bridges—superstructure (I). Taipei, Taiwan: CECI Engineering Consultants, Inc., Taiwan, 2020. ⟨https://www.ceci.org.tw/Upload/Download/2012%E6%B7%B7%E5%87%9D%E5%9C%9F%E6%A9%8B%E6%A2%81%E5%B8%B8%E8%A6%8B%E5%8A%A3%E5%8C%96%E9%A1%9E%E5%9E%8B%E6%8E%A2%E8%A8%8E%E4%B8%8A%E9%83%A8%E7%B5%90%E6%A7%8B(%E4%B8%8A).pdf⟩ (Available).
[16] Chou J-S, Lien J-S, Liu C-Y. Integrative AI and UAV-based visual recognition with metaheuristics for automated repair cost analysis of bridge structural deterioration. Autom Constr 2025;176:106273. https://doi.org/10.1016/j.autcon.2025.106273.
[17] Mundt M, Majumder S, Murali S, Panetsos P, Ramesh V. CODEBRIM Concr DEfect BRidge IMage Dataset 2019. https://doi.org/10.5281/zenodo.2620293.
[18] Kunlamai T, Yamane T, Suganuma M, Chun P-J, Okatani T. Improving visual question answering for bridge inspection by pre-training with external data of image–text pairs. Comput-Aided Civ Infrastruct Eng 2024;39:345–61. https://doi.org/10.1111/mice.13086.
[19] Chun P-J, Chu H, Shitara K, Yamane T, Maemura Y. Implementation of explanatory texts output for bridge damage in a bridge inspection web system. Adv Eng Softw 2024;195:103706. https://doi.org/10.1016/j.advengsoft.2024.103706.
[20] Pu H, Yang X, Li J, Guo R. AutoRepo: a general framework for multimodal LLM-based automated construction reporting. Expert Syst Appl 2024;255:124601. https://doi.org/10.1016/j.eswa.2024.124601.
[21] B. Xiao, H. Wu, W. Xu, X. Dai, H. Hu, Y. Lu, M. Zeng, C. Liu, L. Yuan, Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks, 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4818–4829. Available: https://doi.org/10.48550/arXiv.2311.06242.
[22] L. Beyer *et al.*, PaliGemma: A versatile 3B VLM for transfer, arXiv, vol. abs/2407.07726, 2024, doi: 10.48550/arXiv.2407.07726.
[23] H. Fei, S. Wu, H. Zhang, T.-S. Chua, S. Yan, Vitron: A Unified Pixel-level Vision LLM for Understanding, Generating, Segmenting, editing, arXiv preprint arXiv: 2412.19806 (2024), https://doi.org/10.48550/arXiv.2412.19806.
[24] Huo Y, Ge H, Tang G, Gao S, Xu J. Spiking Trans-YOLO: A range-adaptive energy-efficient bridge between YOLO and Transformer. Neurocomputing 2025;645:130407. https://doi.org/10.1016/j.neucom.2025.130407.
[25] Liu L, Gong H, Zhou Y, Zhou A, Cong L. A YOLO-based network with dynamic feature pyramid for multi-scale bridge surface defect detection. Int J Transp Sci Technol 2025. https://doi.org/10.1016/j.ijtst.2025.01.004.
[26] Ji W, Liu S, Deng L, Li J, Liu Y, Xiong Z. WDI-YOLO: A lightweight steel bridge weld defect detection algorithm using UAV images. J Constr Steel Res 2025;235:109833. https://doi.org/10.1016/j.jcsr.2025.109833.
[27] Yang X, del Rey Castillo E, Zou Y, Wotherspoon L, Yang J, Li H. Automated Concrete Bridge Damage Detection Using an Efficient Vision Transformer-Enhanced Anchor-Free YOLO. Engineering 2025;51:311–26. https://doi.org/10.1016/j.eng.2025.02.018.
[28] Chen S, Cao J, Wan Y, Huang W, Abdel-Aty M. A novel CPO-CNN-LSTM based deep learning approach for multi-time scale deflection basin area prediction in asphalt pavement. Constr Build Mater 2025;458:139540. https://doi.org/10.1016/j.conbuildmat.2024.139540.
[29] Chou J-S, Liu C-Y. Pilgrimage walk optimization: Folk culture-inspired algorithm for identification of bridge deterioration. Autom Constr 2023;155:105055. https://doi.org/10.1016/j.autcon.2023.105055.
[30] Chou J-S, Liu C-Y, Guo P-J. Deploying UAV-based detection of bridge structural deterioration with pilgrimage walk optimization-lite for computer vision. Case Stud Constr Mater 2024;21:e04048. https://doi.org/10.1016/j.cscm.2024.e04048.
[31] Liu C-Y, Chou J-S. Automated legal consulting in construction procurement using metaheuristically optimized large language models. Autom Constr 2025;170:105891. https://doi.org/10.1016/j.autcon.2024.105891.
[32] T. Zhang, V. Kishore, F. Wu, K.Q. Weinberger, Y. Artzi, BERTScore: Evaluating Text Generation with BERT, ArXiv, vol. abs/1904.09675, 2019. Available: https://doi.org/10.48550/arXiv.1904.09675.
[33] J. Davis, M. Goadrich, The Relationship Between Precision-Recall and ROC Curves, in: Proceedings of the 23rd International Conference on Machine Learning (ICML), ACM, Pittsburgh, PA, USA, 2006, pp. 233–240, https://doi.org/10.1145/1143844.1143874.
[34] Berrar D. Cross-Validation. In: Ranganathan S, Gribskov M, Nakai K, Schönbach C, editors. Encyclopedia of Bioinformatics and Computational Biology. Oxford: Academic Press; 2019. p. 542–5. https://doi.org/10.1016/B978-0-12-809633-8.20349-X.
[35] Ting KM. Confusion Matrix. In: Sammut C, Webb GI, editors. Encyclopedia of Machine Learning. Boston, MA: Springer US; 2010. p. 209. https://doi.org/10.1007/978-0-387-30164-8_157.

[36] Deng X, Liu Q, Deng Y, Mahadevan S. An improved method to construct basic probability assignment based on the confusion matrix for classification problem. Inf Sci 2016;340-341:250–61. https://doi.org/10.1016/j.ins.2016.01.033.
[37] Chou J-S, Liu C-Y, Shih H-Y, Lin Z-T. Automated annotation of steel corrosion in UAV-captured images from beneath bridge decks using metaheuristic-optimized computer vision. Structures 2025;75:108696. https://doi.org/10.1016/j.istruc.2025.108696.
[38] A. Assad, M. Arki, M. Sweid, A. Hammad, Crack detection and segmentation for bridges using state-of-the-art deep learning methods: single-stage vs. two-stage detectors, in: Proceedings of the 42nd International Symposium on Automation and Robotics in Construction (ISARC), Montreal, Canada, 2025, p. 996–1003, https://doi.org/10.22260/ISARC2025/0129.
[39] J. Flotzinger, P. Rösch, T. Braml, *dacl10k: Benchmark for Semantic Bridge Damage Segmentation*. arXiv preprint arXiv:2309.00460 (2023), https://doi.org/10.48550/arXiv.2309.00460.
[40] Chun P-J, Yamane T, Maemura Y. A deep learning-based image captioning method to automatically generate comprehensive explanations of bridge damage. Comput-Aided Civ Infrastruct Eng 2022;37(11):1387–401. https://doi.org/10.1111/mice.12793.
[41] C. Chai, Y. Gao, H. Li, G. Xiong, Automatic generation of bridge defect descriptions using image captioning techniques, in: Proceedings of the 10th International Conference on Construction Engineering and Project Management (ICCEPM 2024), Sapporo, Japan, July 29–August 1, 2024, pp. 319–326, https://orca.cardiff.ac.uk/id/eprint/171911/1/ICCEPM2024_paper_82_revised.pdf.